Data Aggregation

Problem Statement: Analyzing Sales Performance by Region in a Retail Company Dataset: "Retail_Sales_Data.csv" Description: The dataset contains information about sales transactions in a retail company. It includes attributes such as transaction date, product category, quantity sold, and sales amount. The goal is to perform data aggregation to analyze the sales performance by region and identify the top-performing regions. Tasks to Perform:

1. Import the "Retail_Sales_Data.csv" dataset.
2. Explore the dataset to understand its structure and content.
3. Identify the relevant variables for aggregating sales data, such as region, sales amount, and product category.
4. Group the sales data by region and calculate the total sales amount for each region.
5. Create bar plots or pie charts to visualize the sales distribution by region.
6. Identify the top-performing regions based on the highest sales amount.
7. Group the sales data by region and product category to calculate the total sales amount for each combination.
8. Create stacked bar plots or grouped bar plots to compare the sales amounts across different regions and product categories.

Code :

1. Import the "Retail_Sales_Data.csv" dataset.

```
import pandas as pd
import numpy as np

df = pd.read_csv('sales_data.csv')

df.head()
```

	date	product	category	price	quantity	revenue	region
0	2022-01-01	Smartphone	Electronics	600.0	10.0	6000.0	East
1	2022-01-01	Laptop	Electronics	1200.0	5.0	6000.0	West
2	2022-01-02	T-Shirt	Clothing	20.0	50.0	1000.0	North
3	2022-01-03	Headphones	Electronics	100.0	20.0	2000.0	East
4	2022-01-04	T-Shirt	Clothing	20.0	25.0	500.0	East

2. Explore the dataset to understand its structure and content.

df.shape

(369, 7)

df.columns

```
Index(['date', 'product', 'category', 'price', 'quantity', 'revenue',
       'region'],
      dtype='object')
```

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 369 entries, 0 to 368
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   date      369 non-null    object
 1   product   369 non-null    object
 2   category  369 non-null    object
 3   price     367 non-null    float64
 4   quantity  368 non-null    float64
 5   revenue   368 non-null    float64
 6   region    369 non-null    object
dtypes: float64(3), object(4)
memory usage: 20.3+ KB
```

df.describe()

	price	quantity	revenue
count	367.000000	368.000000	368.000000
mean	211.226158	14.565217	2060.679348
std	227.335170	8.595740	1910.930790
min	20.000000	3.000000	300.000000
25%	50.000000	8.000000	800.000000
50%	100.000000	12.000000	1200.000000
75%	300.000000	20.000000	2400.000000
max	1200.000000	50.000000	7200.000000

3. Identify the relevant variables for aggregating sales data, such as region, sales

Variable	Purpose
`region`	Grouping data by geographic location
`revenue`	Calculating total sales amount
`category`	Analyzing performance per product type

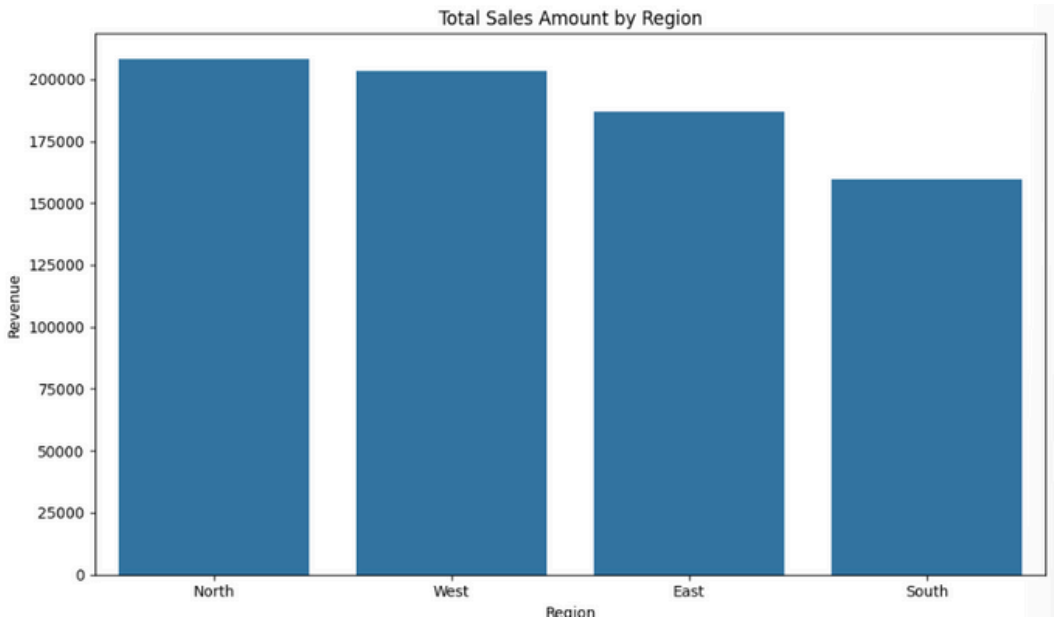
4. Group the sales data by region and calculate the total sales amount for each region.

```
region_sales = df.groupby("region")
["revenue"].sum().reset_index().sort_values(by="revenue", ascending=False)

print(region_sales)
```

```
  region   revenue
1  North  207970.0
3   West  203430.0
0   East  187110.0
2  South  159820.0
```

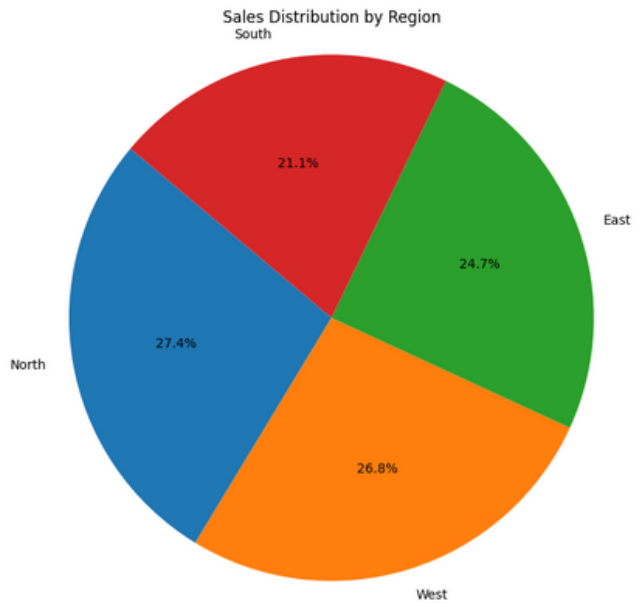
5. Create bar plots or pie charts to visualize the sales distribution by region.

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10, 6))
sns.barplot(x="region", y="revenue", data=region_sales)
plt.title("Total Sales Amount by Region")
plt.xlabel("Region")
plt.ylabel("Revenue")
plt.tight_layout()
plt.show()
```



Pie Chart

```
plt.figure(figsize=(8, 8))
plt.pie(region_sales["revenue"], labels=region_sales["region"], autopct="%1.1f%%",
startangle=140)
plt.title("Sales Distribution by Region")
plt.axis("equal")
plt.show()
```



6. Identify the top-performing regions based on the highest sales amount.

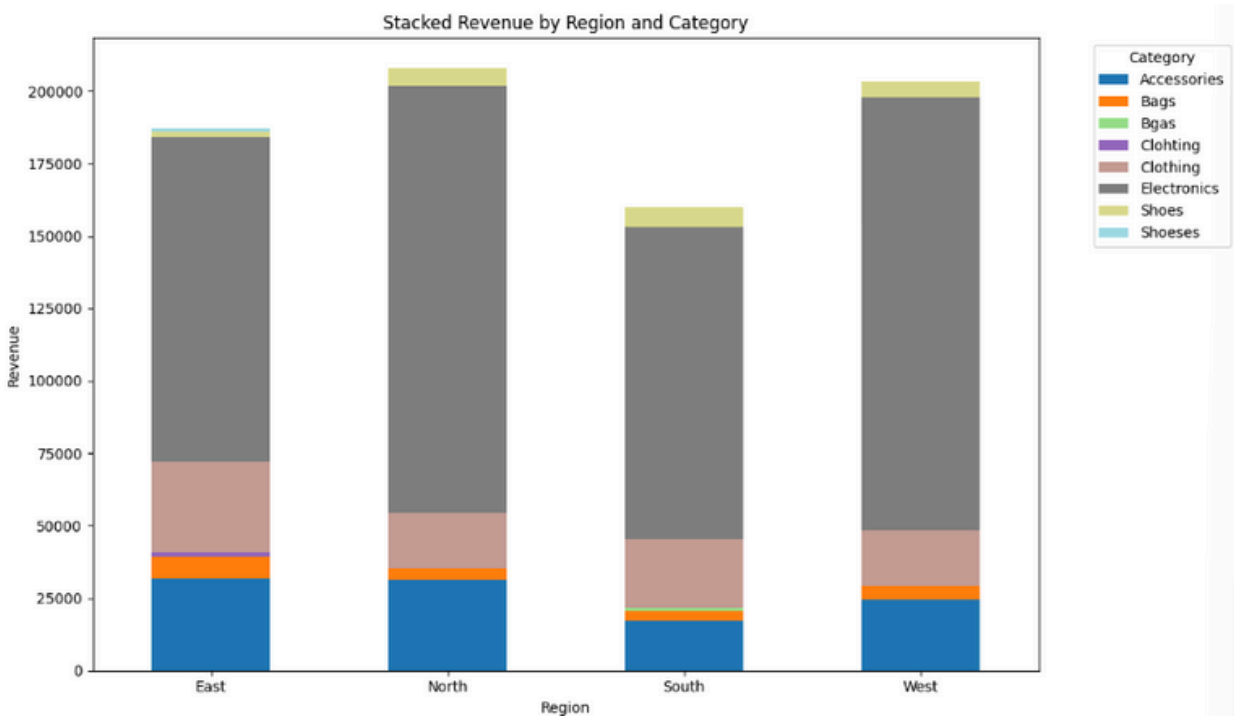
```
top_regions = region_sales.head(3)
print("Top Performing Regions by Sales Amount:")
print(top_regions)
```

```
Top Performing Regions by Sales Amount:
  region   revenue
1  North  207970.0
3   West  203430.0
0   East  187110.0
```

7. Group the sales data by region and product category to calculate the total sales amount for each combination.

```
region_category_sales = df.groupby(["region", "category"])
["revenue"].sum().reset_index().sort_values(by="revenue", ascending=False)

print(region_category_sales.head())
```

```
   region     category   revenue
21   West  Electronics  149400.0
10  North  Electronics  147260.0
4    East  Electronics  111840.0
16  South  Electronics  107580.0
0    East  Accessories   31800.0
```

8. Create stacked bar plots or grouped bar plots to compare the sales amounts across different regions and product categories.

```
# Pivot for stacked bar plot
pivot_stack = region_category_sales.pivot(index="region", columns="category",
values="revenue").fillna(0)

# Plot
pivot_stack.plot(kind="bar", stacked=True, figsize=(12, 7), colormap="tab20")
plt.title("Stacked Revenue by Region and Category")
plt.ylabel("Revenue")
plt.xlabel("Region")
plt.xticks(rotation=0)
plt.legend(title="Category", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```